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Epigenetic age signatures in the forensically relevant body fluid of semen: a preliminary study

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ABSTRACT

To date, DNA methylation has been regarded as the most promising age-predictive biomarker. In support of this, several researchers have reported age predictive models based on the use of blood or even across a broad spectrum of tissues. However, there have been no publications that report epigenetic age signatures from semen, one of the most forensically relevant body fluids. In genome-wide DNA methylation profiles of 36 body fluids including blood, saliva, and semen, the previous age predictive models showed considerable prediction accuracy in blood and saliva but not in semen. Therefore, we selected CpG sites, whose methylation levels are strongly correlated with age in 12 semen profiles obtained from individuals of different ages, and investigated DNA methylation changes at these CpGs in 68 additional semen samples obtained from individuals aged 20 to 73 years using methylation SNaPshot reaction. Among the selected age-related CpG candidates, outstanding age correlation was obtained at cg06304190 in the *TTC7B* gene. Interestingly, the region around the *TTC7B* gene has been reported to show age-related DNA methylation alteration in the sperm methylome of 2 samples collected from individuals at certain time intervals. The age-predictive linear regression model trained with 3 CpGs (cg06304190 in the *TTC7B* gene, cg06979108 in the *NOX4* gene and cg12837463) showed a high correlation between the predicted age and the chronological age, with an average absolute difference of approximately 5 years. These selected epigenetic age signatures are expected to be useful for considerably accurate age estimation in the forensically relevant body fluid of semen. However, because the findings were limited by small sample size, it will be necessary to further evaluate the age correlation of the selected CpGs and to encourage further investigation.

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1. Introduction

Aging is a complex biological process characterized by an overall decline in physiological functions and an increased risk of disease over time [1]. Substantial effort has been devoted to identifying molecular markers that can be used to predict, monitor and provide insights into aging-related physiological decline and disease [2–6]. Moreover, age is an externally visible characteristic that is forensically valuable for predicting an individual's appearance. Therefore, age estimation based on molecular markers is expected to be criminally useful in helping to reduce the number of potential suspects [7,8]. Telomere shortening and accumulation of mitochondrial DNA deletions have been reported to have age dependency [9–10], but they showed low accuracy and various technical problems [4,11]. More recently, a DNA test based on sjTREC DNA quantification has been introduced in the forensic field for chronological age estimation, and this test system achieved R^2 of 0.835 (SE ± 8.9 years) [12]. However, the current most promising age-predictive biomarker is DNA methylation [13–21].

DNA methylation is a major epigenetic modification that occurs at the 5′-position of the pyrimidine ring of cytosine residues within a CpG dinucleotide sequence in adult tissues [22]. Global DNA methylation level decreases with age [23], but many genes or genomic regions have been reported to be hyper- or hypomethylated with increasing age [24,25]. Since the epigenetic landscape varies significantly across tissue types and many age-related DNA methylation changes depend on tissue type [20,25], several previous studies described DNA methylation-based age predictors in specific tissues [13,14,16–19]. For instance, Bocklandt et al. [18] identified 3 CpG sites in the promoters of *EDARADD*,*TOM1L1* and *NPTX2* from saliva using the Illumina HumanMethylation27 (27K) BeadChip array, and built an age predictive model with an average accuracy of 5.2 years. Weidner et al. [14] also identified 3 age-related CpG sites located in the genes *ITGA2B*, *ASPA* and *PDE4C* using the 27K array and subsequent bisulfite sequencing of blood

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samples, and reported a model with a mean absolute deviation (MAD) from chronological age of less than 5 years. Garagnani et al. [17] demonstrated age-related DNA methylation alterations at CpG sites in 3 genes, *ELOVL2, FHL2* and *PENK* with the Illumina HumanMethylation450 (450K) BeadChip analysis of bloods, and suggested *ELOVL2* as the most promising age predictive maker in blood. Later, Zbiec-Piekarska et al. [26] reported an age predictive model for blood using 2 CpGs in the *ELOVL2* gene, which had prediction error of 6.85 years and a MAD from chronological age of 5.03 years. Because the strong age correlation of DNA methylation in the *ELOVL2* gene has been reported in several independent studies [16,17,26–28], *ELOVL2* seems to be one of the most promising age predictive markers in blood to date. On the other hand, age-dependent signatures that were not affected by sex, tissue type or disease state were also reported. A recently reported age-predictive model by Horvath [15] uses 353 CpG sites, and could be applied across a broad spectrum of tissues with a MAD from chronological age of only 3.6 years; Horvath also introduced a freely available online calculator for the epigenetic aging signature. However, age-predictive markers have not yet been defined or evaluated in forensically-relevant body fluid of semen. Even with the model by Horvath, a significant age correlation was not found in sperm, a primary component in semen; the predicted age of sperm was significantly lower than the chronological age of the donor [15].

Meanwhile, an increasing number of rapists are using condoms according to forensic medical examiners' reports. A 1999 study in Oakland, California, found that 13.5% of assailants used condom, probably to protect themselves from identification by DNA profiling (http://www.newscientist.com/article/dn7079). Those condoms are sometimes recovered at the crime scene or at a suspect's home, and can be useful in a police investigation. If DNA profile was obtained from a single source of semen, e.g., from the inside of discarded condom or from the surface of victim's body or from the pellet of differential extraction of sexual assault casework samples, without any alleged suspect, the age estimation with semen will be helpful to reduce the number of potential suspects, thereby contributing to solving crimes.

In the present study, we tested the accuracy of previous age predictive models using genome-wide DNA methylation profiles of 36 body fluids including blood, saliva, and semen. Then, we selected a few age-related CpG candidates from 12 semen profiles obtained through the 450K BeadChip analysis, and tested their age estimation capability by targeted bisulfite sequencing with additional semen samples.

2. Materials and methods

2.1. Samples

Semen samples were collected from 94 healthy male volunteers aged 20 to 73 years using procedures approved by the Institutional Review Board of Severance Hospital, Yonsei University in Seoul, Korea. Among 94 male volunteers, 14 individuals stated that they had a vasectomy. Freshly ejaculated semen was collected in a plastic cup, and 200 μL aliquots of each were stored frozen. DNA was extracted from aliquots of semen using a QIAamp DNA Mini Kit (Qiagen, Hilden, Germany) according to the manufacturer's instructions. Extracted DNA was quantified using a Quantifiler® Duo DNA Quantification Kit (Applied Biosystems, Foster City, CA, USA).

2.2. Age prediction using previously reported age calculators

To test the accuracy of age calculator suggested by Horvath [15], we used the Illumina HumanMethylation450 BeadChip array (Illumina, San Diego, CA, USA) results for forensically relevant body fluids including 12 blood, 12 saliva and 12 semen samples that had been described in our previous report (GSE59505) [29]. DNA methylation data that were downloaded from the Gene Expression Omnibus database (http://www.ncbi.nlm.nih.gov/geo/) were saved as a file with β-values in the comma delimited file (.csv file) format and were uploaded online according to the tutorial for the age calculation of the website (https://dnamage.genetics.ucla.edu/). To make our data comparable to the training data of the epigenetic clock, the default setting of normalize data was used for data submission. Prediction values from the age calculator were received by an e-mail and were compared with the chronological age.

Another age-predictive model suggested by Weidner et al. [14] was also tested with the same dataset of 36 body fluid samples after normalization with a nonparametric empirical Bayes framework method using ComBat within an R package called surrogate variable analysis (http://www.bioconductor.org/packages/release/bioc/html/sva.html) [30,31]. The model uses 3 CpG sites, and age estimation was implemented as follows:

$$\text{Predicted age} = 38.0 - 26.4 \times \text{cg02228185} - 23.7 \times \text{cg25809905} + 164.7 \times \text{cg17861230}$$

2.3. HumanMethylation450 BeadChip data analysis for screening of age-related CpG candidates from semen

To identify age-related CpG candidates from semen, the 450K BeadChip array data obtained from 12 semen samples were analyzed (GSE59505); semen donors' age was 20, 27, 28, 31, 37, 38, 41, 43, 48, 57, 59 and 59 years. Probe sets with signal intensities below the average background for negative control probes (detection *P*-values ≥0.05) were removed from the data set. The calculated β-score corresponds to the percentage methylation value at a specific CpG site and varied between 0 and 1. To adjust β-scores for batch effects among BeadChips, probe sets with missing values were removed, and β-scores were normalized using ComBat [30,31].

To test for the age-association of β-scores at each CpG unit, univariate linear regression was used. CpGs that exhibited a *P*-value <0.01 with an *R*-squared value over 0.7 and an absolute estimate value over 0.005 were considered candidates for age-related CpGs. However, CpGs with probes containing a SNP within 10 bases of the queried site were eliminated.

2.4. Targeted bisulfite sequencing using methylation SNaPshot

The age-relatedness of selected CpG candidates was further investigated using methylation SNaPshot. In consideration of the practicality with regard to the design of bisulfite sequencing PCR primers, a small set of CpG sites were selected from the list of age-related CpG candidates for the subsequent methylation SNaPshot. Methylation SNaPshot based on a single-base extension reaction (SBE) was designed using *in silico*-bisulfite-converted genomic reference sequences as determined by the BeadChip results. PCR primers for the amplification of bisulfite-converted genomic DNA were designed using the Methprimer program (http://www.urogene.org/methprimer/index1.html) [32], and SBE primers for the target CpGs within the PCR products were designed using the Batchprimer3 program (http://wheat.pw.usda.gov/demos/BatchPrimer3/) [33].

PCR was performed in 20 μL reactions containing 1–2 μL of bisulfite-converted DNA, 1.5 U of AmpliTaq Gold® DNA polymerase, 2.0 μL of Gold ST*R 10× Buffer, and 0.4–1.0 μM of each primer

(Table S1). Bisulfite-converted DNA was obtained by modification of 0.5–200 ng genomic DNA using the Imprint® DNA Modification Kit (Sigma–Aldrich Inc., St. Louis, MO, USA) according to the manufacturer's protocol. PCR cycling was conducted in a PTC-200 DNA engine under the following conditions: 95 °C for 11 min; 34 cycles of 94 °C for 20 s, 56 °C for 60 s, and 72 °C for 30 s; and a final extension at 72 °C for 7 min. Then, 5 μL of each PCR product was purified with 1 μL of ExoSAP-IT (USB, Cleveland, OH, USA) by incubation at 37 °C for 45 min followed by heat inactivation at 80 °C for 15 min. SBE was performed using 1 μL of purified PCR product, 0.2–0.4 μM of SBE primer and a SNaPshot™ Kit (Applied Biosystems) according to the manufacturer's instructions (Table S1). Extension products were analyzed using an ABI PRISM 3130 Genetic Analyzer and GeneScan software 3.1 (Applied Biosystems). Percentage methylation value (0–100) at each CpG site was calculated by dividing nucleotide C/G intensity (detection of unconverted methylated DNA) by nucleotide C/G plus nucleotide T/A (detection of converted unmethylated DNA) intensities.

2.5. Age prediction in semen

Initially, the targeted bisulfite sequencing using methylation SNaPshot was performed in 31 normal semen samples. These samples were independent from the BeadChip arrays that were used to select age-related CpG candidates. Based on the methylation SNaPshot results of the initial 31 semen samples, a multivariate linear regression model was generated. During the model construction, stepwise regression was applied with threshold values for F-to-enter of 0.05 and F-to-remove of 0.10. At each step in the stepwise process, the program performs the following calculations: for each variable currently in the model, it computes the t-statistic for its estimated coefficient, squares it, and reports this as its F-to-remove statistic; for each variable not in the model, it computes the t-statistic that its coefficient would have if it were the next variable added, squares it, and reports this as its F-to-enter statistic. At the next step, the program automatically enters the variable with the highest F-to-enter statistic, or removes the variable with the lowest F-to-remove statistic, in accordance with the control parameters we have specified, i.e., F-to-enter of 0.05 and F-to-remove of 0.10. Then, the model generated with relevant CpGs was validated with an independent set of 37 samples. All statistical analyses and data processing were performed using the statistical package IBM SPSS Statistics for Windows, Version 20.0 (IBM Corp., Armonk, NY, USA).

3. Results

3.1. Comparison of previously reported age predictive models using HumanMethylation450 BeadChip array data of 36 body fluid samples

Age prediction values for 36 body fluid samples were obtained using 2 age predictive models and the 450K BeadChip array data. The 36 body fluid samples included 12 blood, 12 saliva and 12 semen samples obtained from a group of volunteers of various ages (age range 20–59 years), and the 2 age predictive models were those suggested by Horvath [15] and Weidner et al. [14].

Although only a limited number of samples were tested, the age calculator suggested by Horvath provided prediction values with considerable accuracy in blood and saliva with a MAD from chronological age of 4.2 and 5.8 years, respectively (Fig. 1 and Table S2). However, the age prediction values for semen were significantly lower than the chronological age of the donors with a MAD from chronological age of 13.3 years, which was consistent with the observation for sperm in a previous report [15].

Contrary to the model by Horvath which uses 353 CpGs, the age predictive model suggested by Weidner et al. uses only 3 CpG sites, and the prediction accuracy was not as high as the authors had described in their report [14]. Calculated MADs from chronological age were over 10 years in all 3 types of body fluids, and it reached to 37.3 years in semen (Fig. 1 and Table S2). These results seemed similar to those reported in a previous article (http://genomebiology.com/2014/15/2/R24/comments), but the analysis has two limitations which are just the same as described in the previous article. First, Weidner et al. mentioned that their predictor works best on pyrosequencing data. Second, Weidner et al. uses a CpG site upstream of cg17861230. Therefore, we further investigated age correlation of the 3 individual CpG sites used in the model, but failed to find close correlation in semen as well as in saliva (Fig. S1).

A recently reported age predictive model for blood also uses only 2 CpGs in the *ELOVL2* gene while demonstrating a low MAD from chronological age of 5.03 years [26]. Because the model used pyrosequencing data for 2 CpG sites adjacent to cg16867657, we could not calculate age predictive values using the model and the array data. Therefore, we investigated age correlation of cg16867657 in 3 types of body fluids, and could observe high correlation in blood ($R^2 > 0.65$) (Fig. S1). These phenomena were also observed at another CpG site, cg06639320 located in the *FHL2* gene, which had been reported to show high age correlation in blood [17] (Fig. S1).

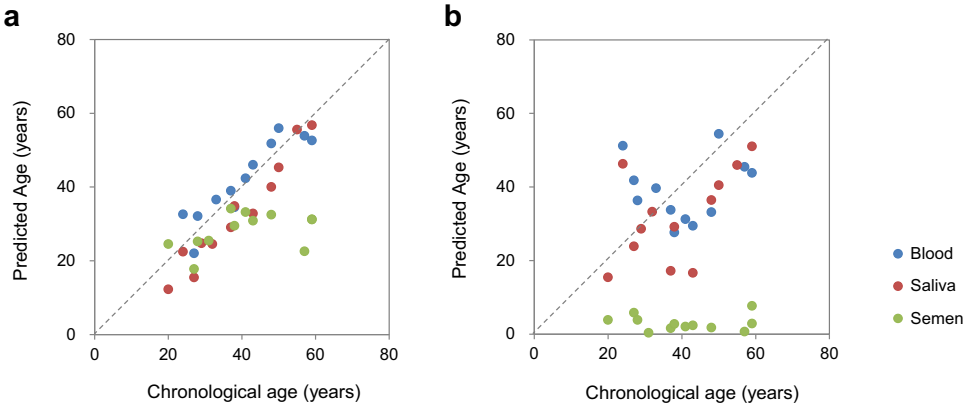


Fig. 1. Comparison of predicted values from age predictive models by Horvath [15] and Weidner et al. [14] using HumanMethylation450 BeadChip array data of 36 body fluid samples. Predicted versus chronological ages of 3 types of body fluids with the model by Horvath (a) and the model by Weidner at al. (b).

Table 1
Targeted bisulfite sequencing results for 24 age-related CpG candidates and their information on Illumina HumanMethylation450 BeadChip analysis.

TargetID	*R* squared ($n=31$)[a]	*P*-value[b]	*R* squared ($n=12$)[b]	Estimate[b]	delta(Max_Min)[b]	CpG island	CHR	MAPINFO_hg19	Gene symbol
cg14750551	0.4656	1.78E-06	0.9071	−0.0052	0.235037064	Shore	3	122401343	PARP14
cg13030797	0.4352	6.21E-06	0.8809	−0.0055	0.265929858	Shore	X	153403150	
cg03063928	0.2063	1.03E-05	0.8684	−0.0050	0.194251452	Other	6	35490164	
cg16256592	0.4201	1.39E-05	0.8604	−0.0060	0.221720174	Shore	5	175664221	C5orf25
cg15477040	0.2801	2.30E-05	0.8458	−0.0063	0.251856521	Shore	17	4124222	ANKFY1
cg23488376	0.4553	2.44E-05	0.8439	−0.0062	0.265592627	Other	19	48367266	
cg00881487	0.4378	3.03E-05	0.8372	−0.0050	0.186184959	Shore	X	3236105	MXRA5
cg16020436	0.3289	3.64E-05	0.8312	−0.0057	0.238086891	Shore	1	203763483	ZC3H11A
cg18989491	0.4331	3.66E-05	0.8311	−0.0058	0.270582094	Shore	10	6185502	PFKFB3
cg12403162	0.3953	4.51E-05	0.8239	−0.0051	0.219676670	Other	10	116394557	ABLIM1
cg13494348	0.4299	4.53E-05	0.8238	−0.0051	0.213673577	Shore	6	170596856	DLL1
cg16856766	0.0791	6.55E-05	0.8106	−0.0064	0.299121774	Other	X	153883245	CTAG2
cg06979108	**0.4418**	6.93E-05	0.8084	0.0055	0.257472781	Other	11	89322851	NOX4
cg02027263	0.0791	9.09E-05	0.7979	−0.0062	0.255500983	Shore	5	175664224	C5orf25
cg04839520	0.3781	9.13E-05	0.7978	−0.0053	0.241931739	Other	7	134922321	STRA8
cg14684375	0.0642	1.20 E-04	0.7867	−0.0071	0.297533487	Island	19	45540895	RELB
cg09340639	0.1914	1.28 E-04	0.7838	0.0050	0.226432069	Other	1	157789662	FCRL1
cg05182393	0.3597	1.37 E-04	0.7811	−0.0058	0.275398234	Shore	8	37821931	ADRB3
cg03818021	0.2829	1.42 E-04	0.7796	−0.0057	0.240811713	Shore	5	1798716	MRPL36
cg12837463	**0.6020**	2.02 E-04	0.7639	−0.0051	0.254960496	Shore	7	35300228	
cg04820254	0.1192	2.10 E-04	0.7620	−0.0053	0.223699236	Shore	20	36147042	BLCAP
cg06304190	**0.6096**	2.36 E-04	0.7566	−0.0056	0.282105006	Shore	14	91283606	TTC7B
cg02337583	0.1659	3.12 E-04	0.7429	−0.0060	0.258223930	Shore	1	203763498	ZC3H11A
cg07813275	0.3716	3.81 E-04	0.7328	−0.0059	0.272018950	Shore	1	182760143	NPL

[a] *R*-squared values obtained from the univariate linear regression of percentage methylation values at a specific CpG site using methylation SNaPshot analysis of 31 semen samples. The *R*-squared values for the CpG sites that had been selected by stepwise regression and subjected to subsequent analysis were indicated in bold.
[b] Values obtained from the univariate linear regression of ß-scores from the beadchip array data for 12 semen samples (GSE59509).

3.2. Selection of age-related CpG candidates from semen

Because age predictive models based on the use of blood or even across a broad spectrum of tissues demonstrated inaccurate age prediction capability in semen, it will be needed to identify novel age-related CpGs for the better prediction of age in semen.

Therefore, we analyzed DNA methylation profiles of 485,000 CpG loci in semen samples collected from 12 individuals aged 20–59 (GSE59509). The number of quality-filtered CpGs was 479,686. To test the association between β-score and age, univariate linear regression analysis was performed separately for each CpG unit. Because the false discovery rate-adjusted *P*-values were all greater than 0.05 except for that of cg21919556, CpGs with *P*-values smaller than 0.01 were initially selected and were considered to be age-related. A total of 10,710 CpGs were selected, and among them, 6,140 CpGs were negatively correlated and 4,570 CpGs were positively correlated with age. Next, CpGs with *R*-squared values higher than 0.7 and absolute estimated values higher than 0.005 were further extracted (Table S3). A total of 106 CpGs were extracted, and the majority of selected CpGs showed a decrease in DNA methylation with age; 94 CpGs were negatively correlated and 12 CpGs were positively correlated with age.

3.3. Selection of CpGs for age prediction in semen

A small set of CpG sites with a high *R*-squared value and a high methylation difference was tested further for age association using methylation SNaPshot. In consideration of the practicality with regard to the design of bisulfite sequencing PCR primers, methylation SNaPshot reaction was designed for 24 CpG sites. Because the results of the methylation SNaPshot strongly correlated with the array data for all samples used in the BeadChip array, the signal intensities for the various nucleotides were used in the calculation of % methylation values without a correction for the differences in signal strength that the different fluorescent labels in the SNaPshot assay have.

Each of the 24 CpG sites was analyzed in independent 31 samples obtained from individuals ranging in age from 24 to 67 years, and their correlation results are shown in Table 1 and Fig. S2. In univariate linear regression analysis, 10 CpGs showed *R*-squared values higher than 0.4, and among them, 2 CpGs (cg06304190 and cg12837463) showed *R*-squared values higher than 0.6.

A multivariate linear regression model using the 24 CpGs showed that these markers explained 95.5% of the variance in 31 males ($R^2 = 0.955$, RMSE = 6.106). However, more than half of the analyzed CpG sites showed *R*-squared values lower than 0.4 and the possibility that multiple CpGs provide redundant information exists. Therefore, stepwise regression, the most popular form of

Table 2
Age-predictive regression models trained with methylation SNaPshot results obtained from training set samples (31 semen samples) and from the total analyzed samples (68 semen samples).[a]

	Target ID	Estimate ($n=31$)[b]	*P*-value	*R*-squared	RMSE[d]	Gene symbol
1	(Intercept)	74.153	0.000	0.814	5.835	
	cg06304190	−0.460	0.000			TTC7B
	cg12837463	−0.353	0.002			
	cg06979108	0.304	0.017			NOX4
	Target ID	Estimate ($n=68$)[c]	*P*-value	*R*-squared	RMSE[d]	Gene symbol
2	(Intercept)	55.357	0.000	0.804	6.466	
	cg06304190	−0.471	0.000			TTC7B
	cg12837463	−0.269	0.002			
	cg06979108	0.491	0.000			NOX4

[a] Age is calculated according to: age = $\beta_0 + \beta_1 \times CpG_1 + \beta_2 \times CpG_2 + \ldots$, where β_0 represents intercept, $\beta_{1,2,3,\ldots}$ represent estimate values of the table, and $CpG_{1,2,3\ldots}$ represent percentage methylation values (0–100) at each target CpG sites as obtained from methylation SNaPshot reactions.
[b] Age-predictor functions obtained from training set samples (31 semen samples).
[c] Age-predictor functions obtained from the total analyzed samples (68 semen samples).
[d] RMSE represents root mean square error of the model.

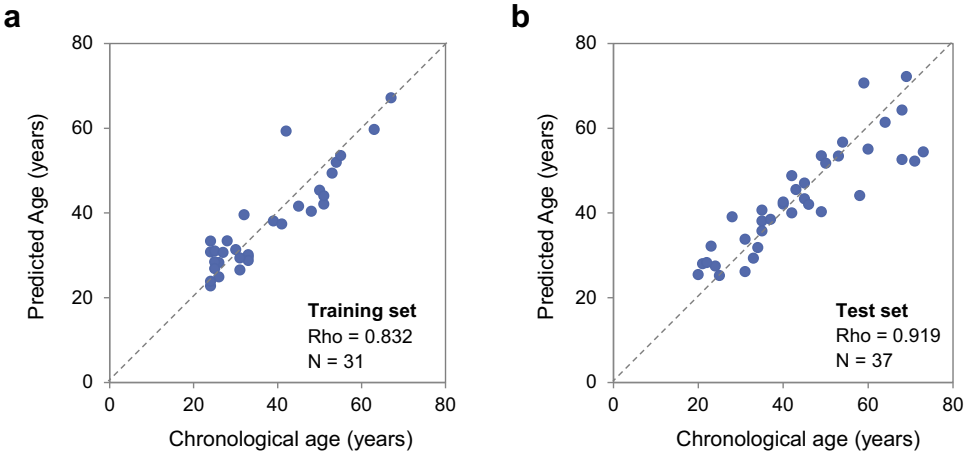


Fig. 2. An age predictive model in semen. DNA methylation percentages at 3 CpG sites (cg06304190, cg12837463 and cg06979108) were modeled using a linear predictor function to produce an age as an outcome in a training set consisting of 31 semen samples. The model was validated using the same regression model in a test set that consisted of 37 semen samples. Predicted versus chronological ages of the training set (a) and the test set (b) of semen.

variable selection, was implemented to select a subset of relevant variables for use in model construction. A stepwise regression analysis produced a model composed of 3 CpGs (cg06304190, cg12837463 and cg06979108), which were all significant variables with high R squared values (0.6096, 0.6020 and 0.4418, respectively). To show the relevance of model, we indicated P-value of each variable in the model (Table 2). This model explained 81.4% of the total variance in 31 males ($R^2 = 0.814$, RMSE = 5.835) (Table 2, Fig. 2a), and showed a high correlation between the predicted and chronological ages (Spearman's Rho = 0.832) with the MAD from chronological ages of 4.2 years.

Subsequently, these 3 CpG loci were further validated in 37 additional semen samples using the same multivariate linear model (Fig. 2b). The MAD from chronological age increased to 5.4 years, but there was still a high correlation between the predicted and chronological ages (Spearman's Rho = 0.919). High deviations from chronological age was frequently observed in advanced aged individuals (>60 years), which may account for aspects of biological age; methylation DNA markers are expected to rather predict biological age than chronological age. However, due to the small sample size, age correlation of markers was not tested separately for young and advanced aged individuals.

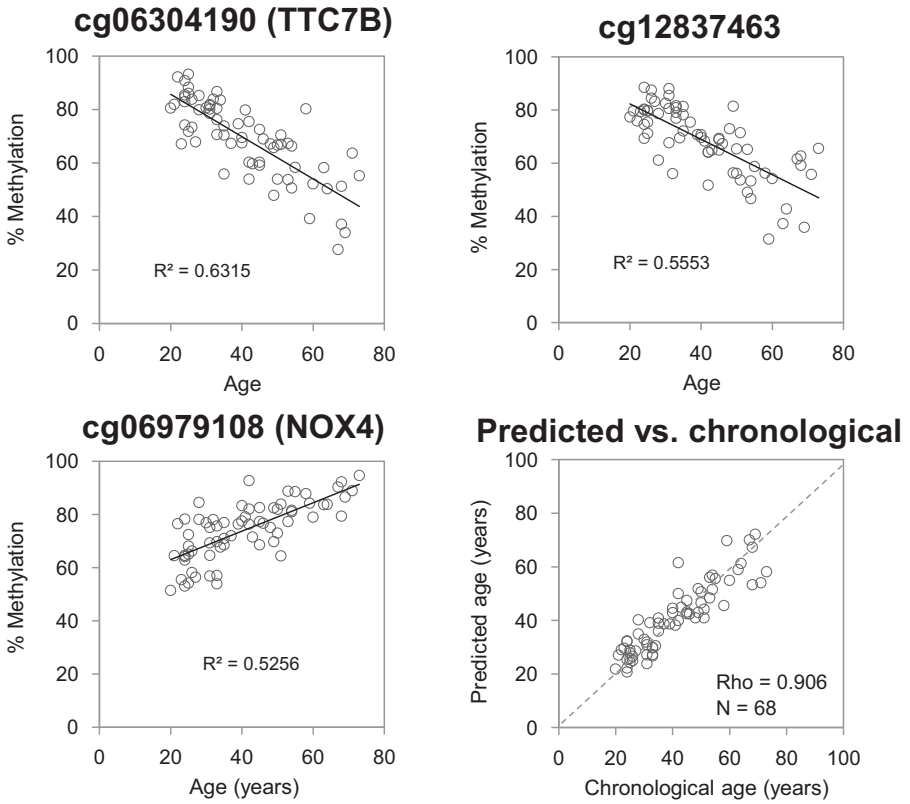


Fig. 3. Correlations between predicted ages and chronological ages of the age-predictive model that was retrained with 68 semen samples. Age correlation of the 3 CpG sites (cg06304190, cg12837463 and cg06979108) and predicted versus chronological ages of 68 semen samples. The MAD from chronological age was 4.7 years (RMSE = 6.466).

4. Discussion

The current study is the first report of an age-predictive model for semen. As expected, age-related CpG sites selected from semen were different from those from blood or other tissues in previous reports [14–18]. Nevertheless, the accuracy of the age-predictive model constructed with 3 age-related CpG sites for semen was comparable to those for blood [14,26]. In addition, a retrained model from a total of 68 semen samples also showed a high accuracy with the MAD from chronological ages of 4.7 years (RMSE = 6.5 years) (Fig. 3 and Table 2).

Among the selected age-related CpGs, the outstanding age correlation was obtained at cg06304190 ($R^2 = 0.6315$) (Fig. 3). Interestingly, cg06304190 is located in the promoter region of the *TTC7B* gene, and the region around the *TTC7B* gene has been reported to show age-related DNA methylation alteration in the sperm methylome of 2 samples collected from individuals 9–19 years apart [34]. Therefore, from age-related CpG candidates with *P*-values smaller than 0.01 in our BeadChip array analysis, we extracted CpG sites located in the regions that the authors of the previous study mentioned to show age-associated methylation alterations, and then investigated DNA methylation changes with age at these CpG sites (Fig. S3). Many CpGs displayed moderate *R*-squared value but low methylation difference. However, the CpG site in the *KCNA7* gene showed relatively high *R*-squared value and high methylation difference, thereby suggesting the possibility that the marker could be another epigenetic age signature in semen.

On the other hand, cg06979108, another relevant variable of the age-predictive model, is located in the upstream region of *NOX4*, and showed positive correlation with age. Like the *TTC7B* gene, the function of Nox4 was unknown until it was recently demonstrated that endothelial-specific expression of the enzyme reduced blood pressure, improved endothelial function and increased angiogenesis in the ischemic hind limb of mice [35]. Therefore, the negative or positive correlation of DNA methylation at *TTC7B* and *NOX4* genes with age suggests the need to further investigate the possible epigenetic regulatory mechanism of these enzymes in terms of aging, which accompanies the decline in physiological functions and increased risk of disease.

By the way, in general, the more age-related CpGs we included in the model, the greater was the age-prediction capability we can obtain. In the comparison of previous models, an age predictor with many CpGs also provided better accuracy than those with a few CpGs. However, the genome-wide DNA methylation analysis requires substantial money, time and effort; therefore, an assay utilizing only a few CpG sites might be more appealing to users if it can provide accuracy comparable to that provided by genome-wide methylation profiling. In addition, because the assay using a small number of CpG sites can be scaled down for relatively small traces of sample, a small set of important age-related CpG sites such as those from Weidner et al. [14] or from our present study could be successfully used in forensic age estimation. Moreover, although the model suggested by Weidner et al. [14] did not work well with array data both from previous and the present studies, it does not prove that the model is inaccurate; the technical limitations in the analyses might partly account for high MAD from chronological ages. Therefore, looked at from that point of view, the age predictive model of the present study could also induce such deviation, and will need a correction when dealing with the data that were not generated using methylation SNaPshot. Nevertheless, since the tendency of age-related CpGs to alter with age will be kept whatever methods are used, the assay for the epigenetic age signatures could be implemented in a variety of ways including pyrosequencing, SBE and NGS of bisulfite converted DNA.

5. Conclusions

Using the data obtained from genome-wide methylation profiling and targeted bisulfite sequencing, we selected epigenetic age signatures from semen. Age-predictive models trained with the 3 selected epigenetic age signatures (cg06304190 in the *TTC7B* gene, cg06979108 in the *NOX4* gene and cg12837463) displayed a high age-predictive capability, with a MAD from chronological age of approximately 5 years, thereby suggesting that the selected epigenetic age signatures could be used for accurate age estimation in semen samples. In addition, as our model uses only a small number of CpG sites and does not require complex bioinformatics, it could be more appealing to researchers and clinicians than the methods which need high-throughput processing. However, because the findings were limited by small sample size, it will be necessary to further evaluate the age correlation of the selected CpGs and to encourage further investigation.

Conflict of interest

The authors declare that they have no conflict of interest.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.fsigen.2015.05.014.

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